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FORAGE — JPMORGAN CHASE & CO. VIRTUAL JOB SIMULATION

# Software Engineering Job Simulation

Internship Presentation & Project Report

INTERNSHIP PERIOD

**July 2026 – August 2026**

## STUDENT CREDENTIALS

**Harshit Bhardwaj**

Enrollment / Roll No.

2410031418

Program & Semester

B.Tech (CSE)  
5th Semester

ORGANIZATION

**Forage — JPMorgan Chase & Co.**

Virtual Software Engineering Job Simulation

# Candidate & Program Overview

## Student Profile

Name: Harshit Bhardwaj  
Roll Number: 2410031418  
Degree: B.Tech (Computer Science & Engineering)  
Academic Batch: 2024–2028  
University: IILM University, Greater Noida, U.P.

## Internship Profile

Program: Software Engineering Job Simulation  
Organization: Forage  
Industry Context: JPMorgan Chase & Co.  
Duration: July 2026 – August 2026  
Certificate of Completion: August 4, 2026

## Project Theme

Hands-on backend engineering focused on scalable system architecture, asynchronous data streaming, lightweight data storage, REST API integration, and controller-layer development.

**The simulation bridges academic concepts with practical, industry-oriented backend implementation.**

# Organization Profile

Forage virtual job simulation platform and the JPMorgan Chase & Co. engineering context

## Forage

A platform dedicated to inspiring and empowering future professionals through virtual job simulations.

## Simulation Context

The tasks were modeled after real-world engineering challenges involving robust, scalable financial technology solutions.

## Industry Relevance

Enterprise applications need backend services that are reliable, scalable, decoupled, and capable of handling real-time data.

## Learning Environment

The simulation provided practical task parameters and a structured environment for applying software engineering concepts.

## Core Focus

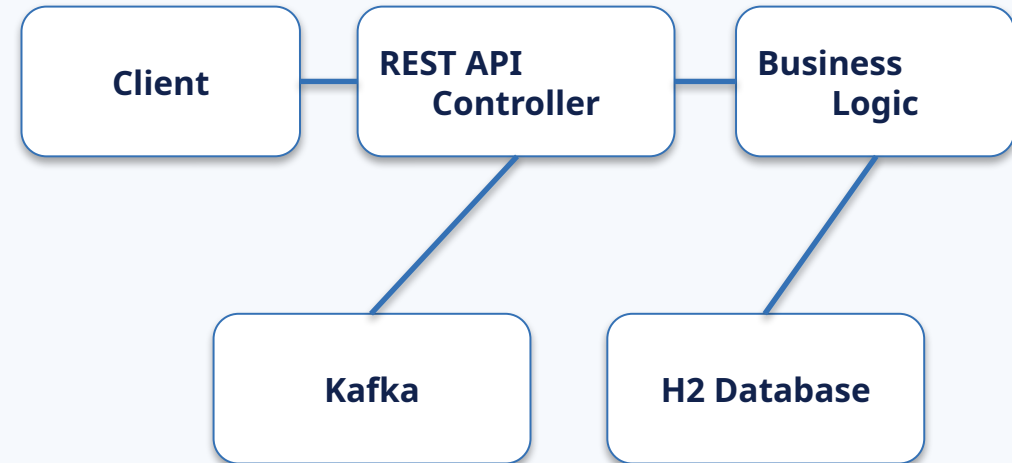
Backend architecture, event streaming, database integration, RESTful communication, and HTTP request handling.

# Problem Statement & Industry Context

## The Industry Challenge

- Modern enterprise applications process large volumes of data in real time.
- Backend systems must be scalable, decoupled, robust, and responsive.
- The simulation required asynchronous messaging and lightweight data storage.
- These capabilities needed to be exposed through a standardized web interface.

### Target Backend Flow



**Objective: construct a foundational backend architecture that can handle asynchronous messaging, transient storage, and standardized HTTP access.**

# Project Objectives

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## 01 Project Setup

Establish the development environment, configure dependencies, and initialize the scalable backend codebase.

## 02 Kafka Integration

Implement message-broker based asynchronous data streaming and decoupled component communication.

## 03 H2 Integration

Use an H2 in-memory database for rapid testing, application storage, and efficient query execution.

## 04 REST API Integration

Connect internal backend services through RESTful architectural principles.

## 05 REST API Controller

Build the controller layer to handle HTTP requests, orchestrate backend logic, and return responses.

# Technologies & Tools Used

## Messaging & Event Streaming

Apache Kafka

- Producers and consumers
- Asynchronous data handling
- Decoupled communication

## Database Management

H2 In-Memory Database

- Lightweight relational storage
- Fast runtime persistence
- CRUD operations

## Web Architecture

RESTful APIs

- Stateless client-server communication
- HTTP methods: GET, POST, PUT, DELETE
- Controller-based request handling

## Technical Focus

Project Setup

Kafka Integration

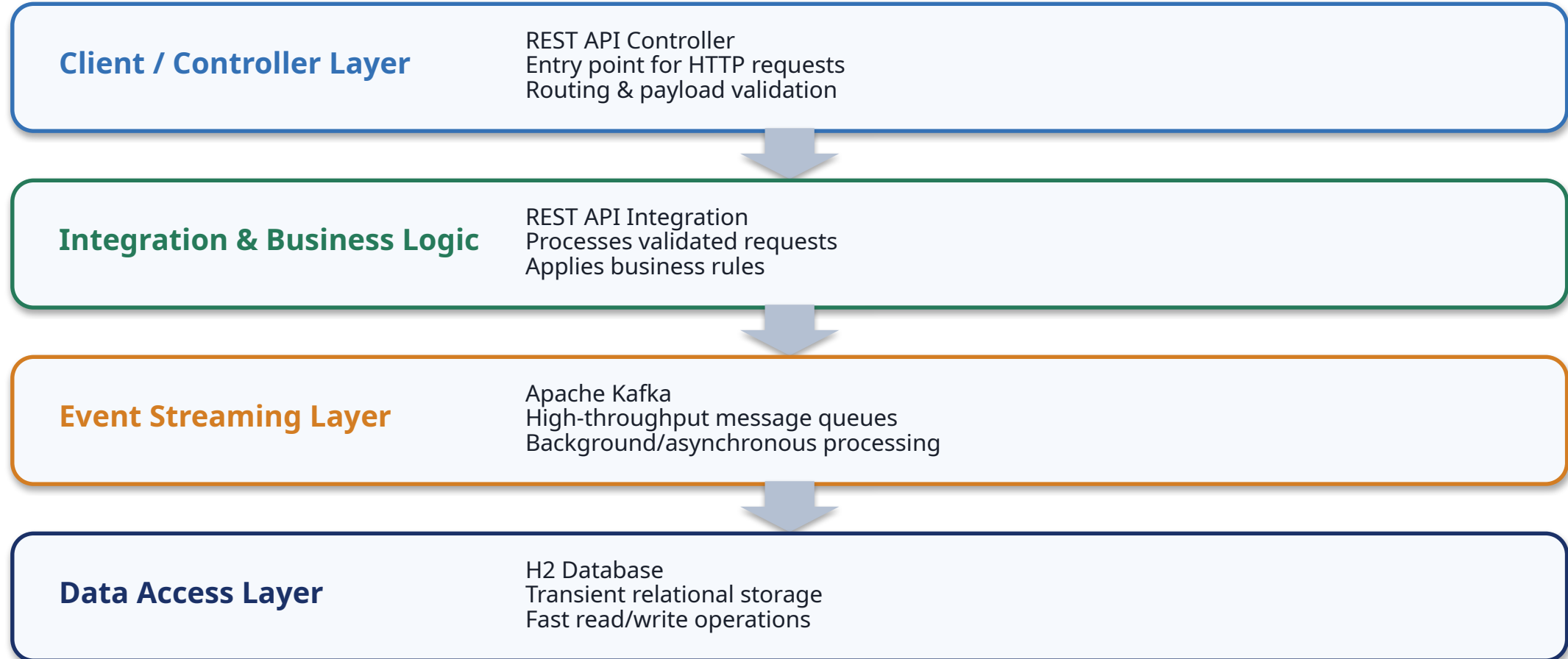
H2 Integration

REST API Integration

REST API Controller

# System Architecture

Multi-tier backend architecture designed for separation of concerns



Separation of concerns enables clearer data flow, easier testing, and modular backend integration.

# Methodology & Execution

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Sequential, iterative execution modeled around practical engineering phases.



**Outcome: disparate backend services were tied together and exposed through functional REST controller endpoints.**



# Project Scope & Deliverables

## Included in Scope

- Backend software development
- System integration
- Data flow management
- H2 transient database configuration
- Kafka event streaming setup
- REST API integration
- REST API Controller layer

## Deliberately Outside Scope

- Front-end development
- User Interface (UI) design
- User Experience (UX) optimization
- Cloud deployment pipelines / CI-CD

This maintained a strict focus on core backend engineering.

**The scope covered the backend data pipeline from transient storage and event streaming through API exposure.**

# Expected Outcomes & Skills Demonstrated

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- 1 Backend Foundations** Project configuration, dependency management, and scalable backend initialization.
- 2 Event-Driven Processing** Kafka-based production and consumption of asynchronous data streams.
- 3 Data Persistence** H2-backed transient storage with efficient in-memory CRUD operations.
- 4 API Engineering** RESTful communication and controller-level HTTP request/response handling.
- 5 System Integration** Coordination of messaging, storage, business logic, and web interfaces.

**Overall result: a functional backend architecture integrating APIs, messaging, and transient data storage.**

# Internship Completion Certificate

Certificate issued for the Software Engineering Job Simulation

## 3. INTERNSHIP COMPLETION CERTIFICATE

JPMORGAN CHASE & CO.



Inspiring and empowering  
future professionals

## Harshit Bhardwaj Software Engineering Job Simulation

Certificate of Completion  
August 4th, 2026

**Certificate of Completion — August 4, 2026**

Over the period of July 2026 to August 2026, Harshit Bhardwaj has completed the following:

Project Setup  
Kafka Integration  
H2 Integration  
REST API Integration  
REST API Controller

A handwritten signature in black ink, appearing to read "Tom Brunskill".

**Tom Brunskill**  
CEO, Co-Founder of  
Forage

# References

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- Forage & JPMorgan Chase & Co., "Software Engineering Job Simulation," Certificate of Completion, August 4, 2026.
- Apache Software Foundation, "Apache Kafka Documentation."
- H2 Database Engine, "H2 Database Engine Documentation."
- Spring Framework, "Spring Boot Web (REST API) Reference Guide."
- OpenAPI Initiative, "OpenAPI Specification for REST API Documentation."

## Source Basis

Project details and completion information are grounded in the supplied internship report and reference presentation.

# Thank You!

**Harshit Bhardwaj**

Roll No: 2410031418

B.Tech (CSE) | 5th Semester

School of Computer Science & Engineering

IILM University, Greater Noida, U.P.